

Vemuri Amrutha

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Objective

Electronics and Communication Engineering student with hands-on experience in embedded systems, IoT prototyping, and hardware-software integration using ESP32, STM32, Raspberry Pi, and Arduino. Currently interning at **DRDO – Research Centre Imarat (RCI), Hyderabad** on a physical embedded systems project. Proficient in C, Python, sensor integration, UART/I2C/SPI protocols, and real-time automation. Seeking an IoT Engineer Internship to contribute to commercial-grade IoT product development.

Internship Experience

- Embedded Systems Intern** (On-site) June 10, 2026 – Present
Defence Research & Development Organisation – RCI (DRDO-RCI) Hyderabad, India
- Working on embedded systems development in a physical lab environment at **Research Centre Imarat**, a premier DRDO lab specializing in missile systems and avionics.
 - Gaining hands-on exposure to hardware-level programming, sensor interfacing, and real-time system design in a mission-critical, defence-grade setting.
 - Collaborating with senior engineers on embedded hardware prototyping, testing, and system-level validation under structured R&D protocols.

Education

- Vignan Institute of Technology and Science** Hyderabad, India
B.Tech – Electronics and Communication Engineering 2023 – 2027
- CGPA: **8.51 / 10.0**
- SR Junior College** Hyderabad, India
Intermediate – Mathematics, Physics, Chemistry (MPC) 2021 – 2023
- Percentage: **82.8%**

Project Experience

- Multi-Language Voice-Controlled IoT Home Automation** | *Python, Raspberry Pi, Google Assistant*
- Built a smart home automation system enabling multilingual voice command control of appliances via Google Assistant integrated with Raspberry Pi GPIO-based relay switching.
 - Developed Python backend automation logic for remote appliance management, implementing end-to-end hardware-software integration and real-time device control – directly aligned with smart home IoT product development.
- Smart Waste Management System** | *ESP32, Ultrasonic Sensor, Embedded C*
- Designed an ESP32-based smart dustbin prototype using ultrasonic proximity sensing for automated touchless lid actuation and real-time fill-level monitoring.
 - Integrated sensor input processing with embedded control logic for responsive, low-cost automation – demonstrating skills applicable to ESP32-based commercial IoT product development.
- Smart Door Lock System** | *Arduino Uno, Fingerprint Sensor, Embedded C*
- Developed a biometric door lock system using Arduino Uno and a fingerprint sensor module for secure, touchless identity-based access control.
 - Implemented fingerprint enrollment, authentication logic, and relay-driven locking mechanism – demonstrating practical embedded hardware-software integration for security IoT applications.

Technical Skills

Languages : C, Embedded C, Python (*Intermediate*), Java (*Basic*)

IoT & Embedded Platforms : ESP32, STM32 (*Familiar*), Raspberry Pi, Arduino Uno

Embedded Concepts : GPIO Interfacing, Sensor Integration, Relay Control, Fingerprint Sensor Interfacing, Real-Time Control, Automation Systems, Interrupt Handling, OTA Updates

Communication Protocols : Wi-Fi (ESP32), UART, I2C, SPI, MQTT (*Basics*)

Cloud & Connectivity : Mobile App Connectivity (REST APIs), Cloud Dashboard Integration, IoT Data Pipelines

Core Subjects : Digital Electronics, Microprocessors & Microcontrollers, Verilog HDL, Signals & Systems, EDC, IoT

Development Tools : Git, GitHub, VS Code, Proteus, Multisim, Scilab

Soft Skills : Cross-Functional Collaboration, Technical Articulation, Analytical Reasoning, Learning Agility

Relevant Coursework

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|--------------------------------------|---------------------------------|
| • Digital Electronics | • Signals & Systems |
| • Embedded Systems | • Analog Communications |
| • Internet of Things (IoT) | • Electronic Devices & Circuits |
| • Microprocessors & Microcontrollers | • Verilog / HDL Basics |